MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 6164

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: TR, bandpass, controlled phase shift, frequency range 8,500 to 9,600 MHz

ABSOLUTE RATINGS:

Parameter:	Incident power	Ebb	prr	Du	Alt
Unit:	kw	Vdc	pps	---	ft
Maximum:	---	---	1,000	0.001	10,000
Minimum:	4	-750	---	---	---

PHYSICAL CHARACTERISTICS: See figure 1

Application: See note 1

TEST CONDITIONS:

Parameter:	Incident power	tp1	tp2	prr	Ebb	Ri	σ'	F
Unit:	kw	µs	µs	pps	Vdc	Meg	---	MHz
Tolerance:	±10%	±0.15	±0.15	---	±25	---	max	±6%
Test condition 1:	40	1.0	0.5	1,000	-625	3.3	1.10	F3
Test condition 2:	200	1.0	---	1,000	-625	3.3	---	F3

GENERAL:

First article sample approval - Required (see note 2)

FREQUENCY	
F	MHz
1	8,500
2	8,775
3	9,050
4	9,325
5	9,600

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:59Z
Check the source to verify that this is the current version before use.

6. / Changes from previous issue. Marginal notations are not used in this

METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	3						
1027	Temperature cycling (nonoperating)	---	---	10 cycles (minimum)	---	---	---	---
4406	Ignitor voltage drop	---	---	Ii = 100 µAdc	Eid	200	375	Vdc
4416	Insertion loss	---	---	F = F3 ±50 MHz; Ri = 3.3 Meg; Ebb = -625 ±25 Vdc	Li	---	1.0	dB
4452	Flat-leakage power	---	1		pF	---	30	mw
4452	Spike-leakage energy	---	1		Ws	---	0.15	erg
4494	Position of short	4	1	F3 ±10 MHz	Distance	---	±0.007	inch
---	Reflection difference	5	---	F1 thru F5; Ebb = -625 ±25 Vdc; Ri = 3.3 Meg	SWR	---	2	dB
---	Transmission difference	5	---	F1 thru F5; Ebb = -625 ±25 Vdc; Ri = 3.3 Meg	Phase	---	5	degrees
	Quality conformance inspection, part 2							
4021	Degradation due to vibration	---	---		---	---	---	---
4101	Dielectric material strain	---	---		---	---	---	---
4401	Ignitor ignition time	---	---	Ebb = -625 ±25 Vdc; Ri = 3.3 Meg	t	---	5	sec
4471	Recovery time	---	2		t	---	10	µs
4488	Arc loss	---	1	po = 4 kw ±10%	La	---	0.8	dB
	Quality conformance inspection, part 3							
---	Life tests	---	2	Group D	t	500	---	hrs
---	Life-test end points:							
4416	Insertion loss	---	---	F = F3 ±50 MHz; Ri = 3.3 Meg; Ebb = -625 ±25 Vdc	Li	---	1.2	dB
4452	Flat-leakage power	---	1		pF	---	30	mw
4452	Spike-leakage energy	---	1		Ws	---	0.15	erg
4471	Recovery time	---	2		t	---	15	µs
1027	Temperature cycling life	---	---	Group C; 12 cycles (minimum)	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:59Z
Check the source to verify that this is the current version before use.

NOTE

1. The type 6164 is an integral cavity broad-band transmit-receive tube designed for use in the RG-51/U waveguide. The magnitude of the vector difference between tubes of low-level reflection and transmission are not greater than 4 dB SWR and 10°, respectively, over the range of 8,500 to 9,500 MHz.

2. First article inspection shall consist of performing all tests listed on this tube specification sheet. The sample size and allowable defects shall be in accordance with MIL-E-1, Qualification, Samples, and acceptance criteria. Three copies of the test report, prepared in accordance with MIL-STD-813, shall be forwarded to the purchasing activity for evaluation by the preparing activity.

   Invitation for bids should provide that the preparing activity reserves the right to waive the requirements for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

3. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

4. With a metal plate shorting the line, the position of the VSW minimum shall be determined. The metal plate shall be replaced by the TUT and the position of the VSW minimum of the flat position of the pulse shall be measured and shall be 0.032 inch (0.81 mm) further from the magnetron within the limits specified.

5. This test shall be conducted in accordance with test procedure for reflection and transmission differences described herein.

6. Revision letter symbols are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 17, 80
DSA - ES
NS

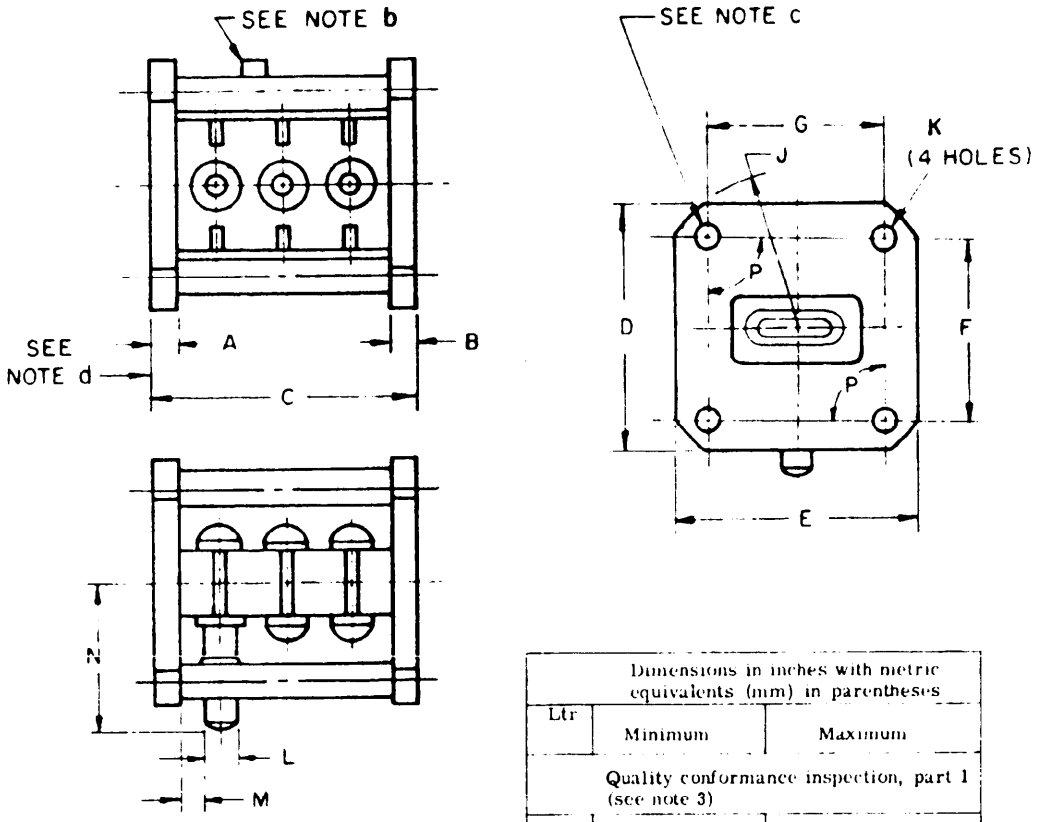
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2843)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:59Z
Check the source to verify that this is the current version before use.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
	Quality conformance inspection, part 1 (see note 3)	
A	.190 (4.83)	.220 (5.59)
B	.190 (4.83)	.220 (5.59)
C	1.973 (50.11)	1.993 (50.62)
D	1.844 (46.84)	1.906 (48.41)
E	1.844 (46.84)	1.906 (48.41)
J	.438 (11.13)	.500 (12.70)
K	.169 (4.29) Dia. (see note c)	
L	.245 (6.22)	.255 (6.48) Dia.
M	.156 (3.96)	---
N	---	.594 (15.09)
	Quality conformance inspection, part 2	
F	1.468 (37.29)	1.480 (37.59)
G	1.356 (34.44)	1.368 (34.75)
P	89.5	90.5°

NOTES:
a. Silver plate 100 MSI, or equivalent.
b. Exhaust tube shall not extend beyond flange more than .25 (6.35 mm).
c. The rectangle formed by the centers of the four .169 Dia. holes shall be centered on the flange face within .031 (.794 mm).
d. The planes of the two waveguide-flange surfaces shall be parallel within ±.002, max.

FIGURE 1. Outline drawing of electron tube type 6164.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:59Z
Check the source to verify that this is the current version before use.

TEST PROCEDURE

Reflection and transmission differences

1. Purpose. The purpose of this test is to measure the reflection and transmission differences.

2. Procedure. The reflection and transmission differences shall be measured, with ignitor currents in the range specified, by comparing the TUT with a standard dummy at F1, F2, F3, F4, and F5, in a special waveguide bridge circuit. The testing procedure and equipment used in testing the TUT shall be as follows:

2.1 Tuning. The tuning and low-level acceptance testing of the tube shall be accomplished by comparing the tube with a comparison-standard dummy (see 2.6), in the waveguide bridge circuit (see 2.5). The TUT and the comparison-standard dummy may be inserted in opposite branches of the waveguide bridge circuit in the test openings designated 1 and 2.

2.2 Signal. The signal source and difference measuring display shall comprise a continuously-swept oscillator covering the frequency band from 8,500 to 9,600 MHz, synchronized with a dual-channel oscillographic display (Polarad Model I, or equivalent). This signal display shall consist of two signals proportional to the magnitude of the vector differences in reflection and transmission between the tube under test and the comparison-standard dummy, as obtained from the detectors designated R and T in the waveguide bridge circuit.

2.3 Maximum difference. This specification specifies the maximum magnitude of the vector differences in low-level reflection and transmission between each tube and the comparison-standard dummy. The values of these vector differences are as follows:

2.3.1 2 dB SWR in reflection.

2.3.2 5 degrees in transmission.

2.4 Acceptance limits. The acceptance limits specified in the specification sheet and covered above in 2.3 shall be set by comparing each of the differential standards described below in 2.7.1 and 2.7.2 with the precision guide section described below in 2.7.3. The precision guide section shall be inserted in one of the test openings and the differential standards inserted successively in the other test opening shown in figure B. The magnitude of each of the resulting display deflections shall be adjusted to correspond with limit lines on the scope fact by adjusting the gain controls of each display circuit. The setting of waveguide attenuator P_1 shall be determined experimentally beforehand by adjusting it for the maximum amount of attenuation which permits both steps of the calibration procedure described above to be performed without readjusting the attenuator or without excessive noise on the display; a minimum attenuation of 6 dB is required to pad the oscillator from the bridge circuit.

2.5 Waveguide bridge circuit. The waveguide bridge circuit is shown in figure B, and shall meet the following requirements:

2.5.1 Mechanical. The mechanical requirements of the waveguide bridge circuit are as follows:

2.5.1.1 The waveguide parts shall have the inner dimensions of RG-51/U waveguide (1.497 x 1.122 inches) (38.02 x 28.50 mm).

2.5.1.2 Both waveguides at the test openings 1 and 2 shall be terminated by UG-52A/U flanges.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:59Z
Check the source to verify that this is the current version before use.

2.5.1.3 Each pair of flanges at test openings 1 and 2 shall be fastened rigidly together to maintain the waveguides parallel and alined, and the openings 2.000 +.002 inches long (50.80 ±0.05 mm).

2.5.1.4 Test parts in openings 1 and 2 shall be held to flanges nearest signal input by even pressure applied at four points at opposite end.

2.5.2 Electrical. The electrical requirements of the waveguide bridge circuit are as follows:

2.5.2.1 Over the band from 8,500 to 9,600 MHz, the reflection measured in either direction at the planes shall be as indicated below:

Plane	Maximum SWR
A	less than 3 dB
B	less than 2 dB
C	less than 3 dB
D	less than 2 dB

2.5.2.2 Over the band, the input to pads P_2 and P_3, with precision guide sections figure G in test openings 1 and 2, shall be down 35 dB from the signal input at plane A.

2.5.2.3 Waveguide pads P_2 and P_3 shall introduce at least 10 dB of attenuation. Pad P_1 is adjustable with a minimum attenuation of 6 dB and is set as described above in 2.4, acceptance limits.

2.6 Comparison-standard dummy. The comparison-standard dummy shall be similar to that shown in figure A and shall meet the following electrical requirements:

2.6.1 Reflection. The reflection at 12 frequencies of the comparison-standard dummy shall fall within a radius of 0.5 dB SWR. (1.06 VSWR measured at the center of a Smith Chart) of the following locations on a Smith Chart when measured as described:

Frequency (+S MHz)	Location: Resistance ($\frac{R}{Zo}$)	Location: Reactance ($j\frac{X}{Zo}$)
8,500	1.19	-.27
8,600	1.08	+.12
8,700	.91	-.02
8,800	.73	+.10
8,900	.82	+.24
9,000	1.01	+.12
9,100	.91	-.12
9,200	.73	-.09
9,300	.71	+.06
9,400	.88	+.16
9,500	1.03	+.10
9,600	1.27	+.18

These measurements shall be made in RG-51/U waveguide with the dummy followed by a termination which has a reflection of less than 0.1 dB SWR. The reference for the measurement shall be at the input flange of the dummy, marked IN, corresponding to the high-power end of a 6164 TR tube.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:59Z
Check the source to verify that this is the current version before use.

2.6.2 Transmission. The transmission properties of the comparison-standard dummy shall be tested by determining the frequencies of resonance in the circuit shown in figure C with the fittings shown in figure D. For this test, it shall be possible to measure the frequency to within ±1 MHz. The minimum loss in transmission through the comparison-standard dummy shall occur at the following frequencies:

8,394 ± 5 MHz
8,556 ± 5 MHz
8,966 ± 5 MHz
9,433 ± 5 MHz
9,705 ± 5 MHz

2.6.3 Stability. The comparison-standard dummy shall meet all the requirements specified in 2.6, 2.6.1, and 2.6.2 after temperature cycling over the range of -50° to +100°C.

2.7 Differential standards.

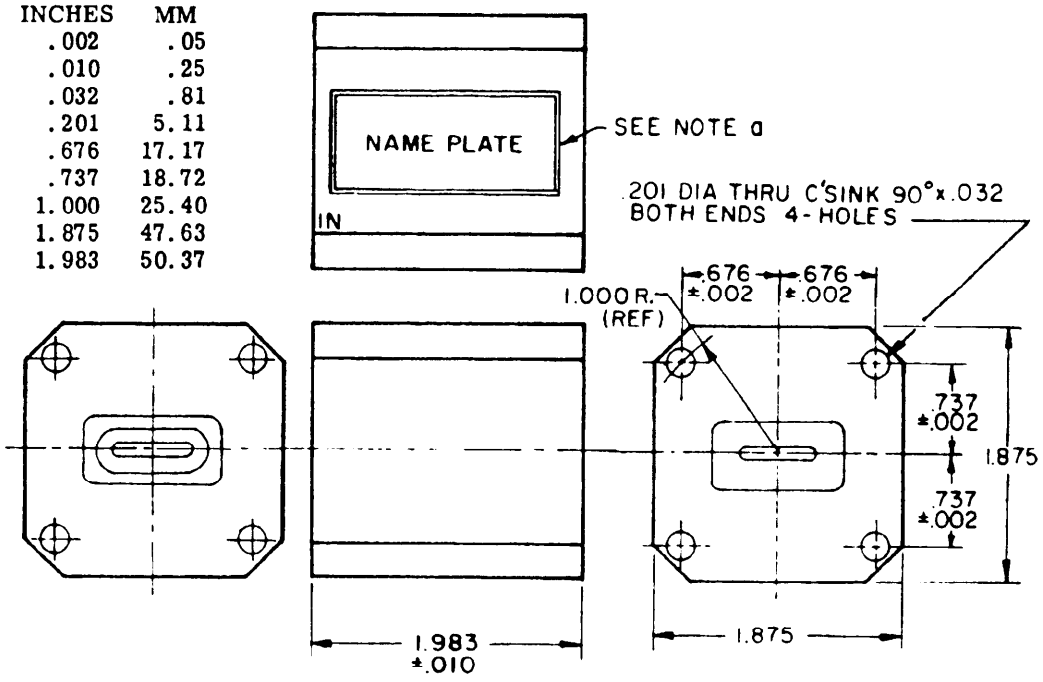
2.7.1 Reflection. The differential reflection standard used to determine the reflection acceptance limits of 2.0 dB SWR shall be as shown in figure E.

2.7.2 Transmission. The differential transmission standard used to determine the transmission acceptance limits of 5.0 degrees shall be as shown in figure F. An alternate design such as the Wheeler Laboratories Model 261, or equivalent is also acceptable.

2.7.3 Precision guide. The precision guide used in the waveguide comparison circuit with the differential standards shall be as shown in figure G.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:59Z
Check the source to verify that this is the current version before use.

INCHES	MM
.002	.05
.010	.25
.032	.81
.201	5.11
.676	17.17
.737	18.72
1.000	25.40
1.875	47.63
1.983	50.37



NOTES:

a. The nameplate shall give the following information:
   (1) The organization which tuned and tested the tube.
   (2) The model and serial number of the tube.
   (3) The date the tube was completed and sealed.

b. The electrical performance of the tube shall conform to tube specification notes 2.6, 2.6.1, 2.6.2, and 2.6.3.

FIGURE A. Comparison - standard dummy tube.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:59Z
Check the source to verify that this is the current version before use.

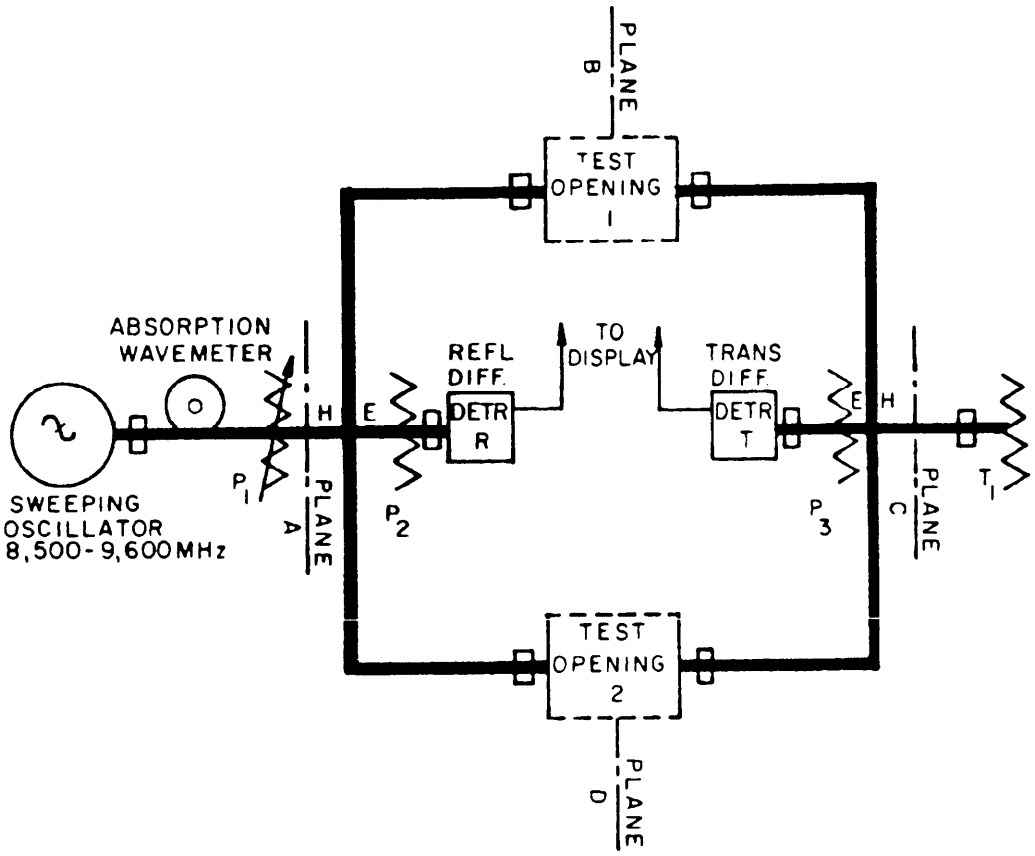


FIGURE B. Waveguide bridge circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:59Z
Check the source to verify that this is the current version before use.

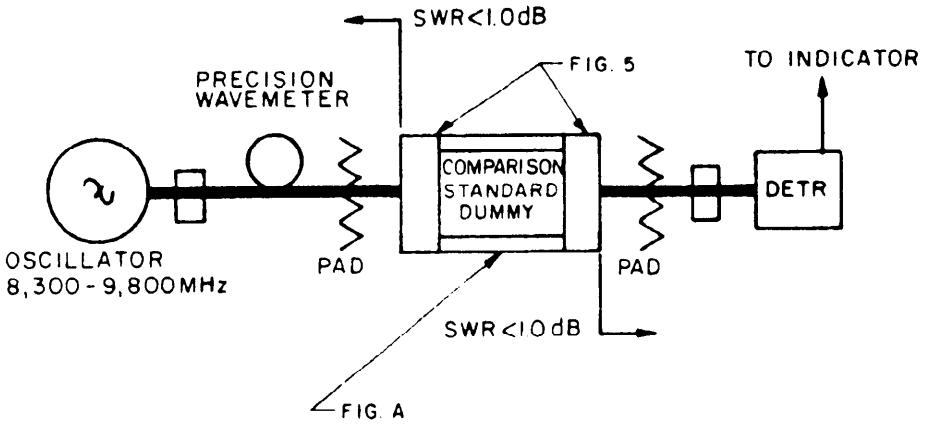
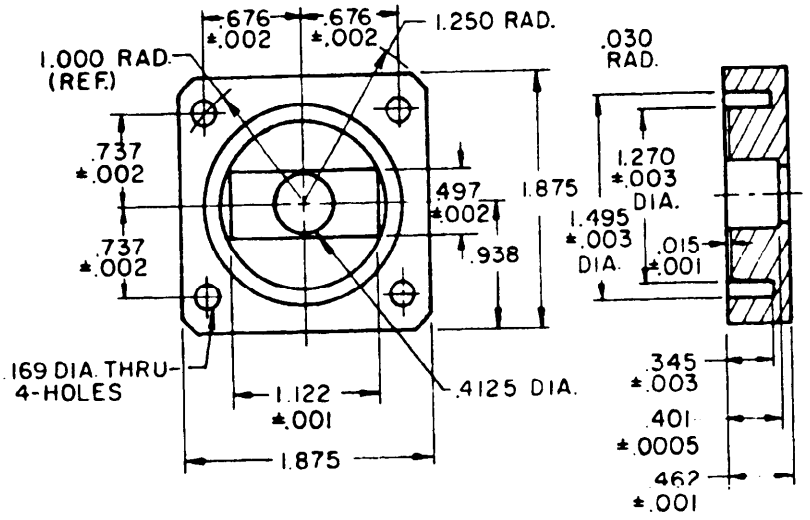


FIGURE C. TR resonance test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:59Z
Check the source to verify that this is the current version before use.



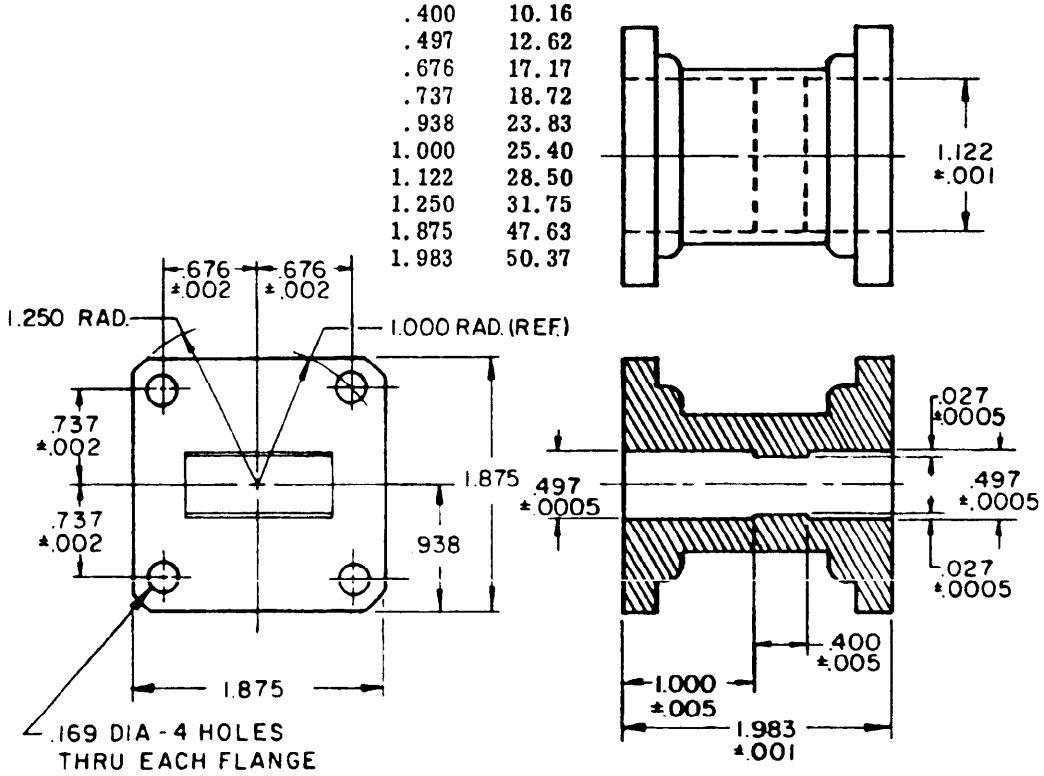
INCHES	MM
.0005	.01
.001	.03
.002	.05
.003	.08
.015	.38
.030	.76
.169	4.29
.345	8.76
.401	10.19
.4125	10.48
.462	11.73
.497	12.62
.676	17.17
.737	18.72
.938	23.83
1.000	25.40
1.122	28.50
1.250	31.75
1.270	32.26
1.495	37.97
1.875	47.63

NOTES:
a. Part can be built up from UG-52A/U flange.
b. Tolerance on all decimal dimensions to be ±.005 unless otherwise specified.

FIGURE D. TR tube test part.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:59Z
Check the source to verify that this is the current version before use.

INCHES	MM
.0005	.01
.001	.03
.002	.05
.005	.13
.027	.69
.169	4.29
.400	10.16
.497	12.62
.676	17.17
.737	18.72
.938	23.83
1.000	25.40
1.122	28.50
1.250	31.75
1.875	47.63
1.983	50.37

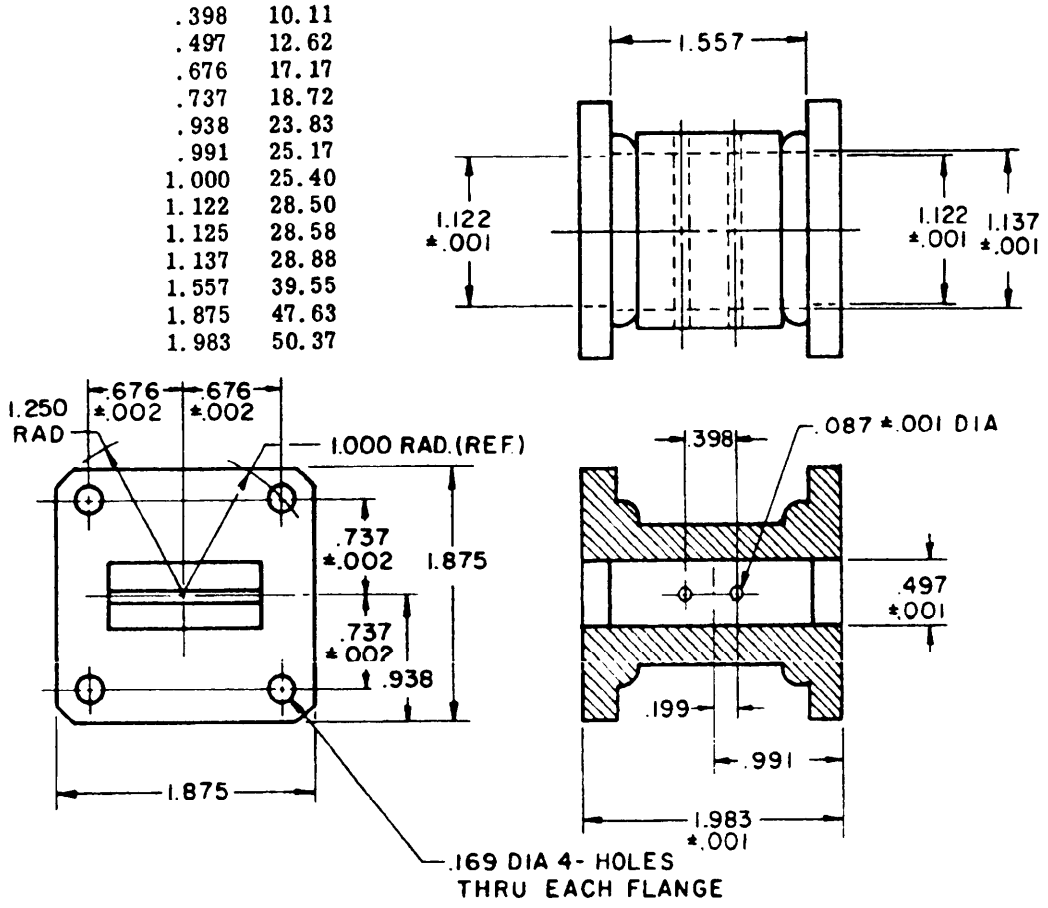


NOTE: Tolerance on all decimal dimensions to be ±.005 unless otherwise specified.

FIGURE E. Differential reflection standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:59Z
Check the source to verify that this is the current version before use.

INCHES	MM
.001	.03
.002	.05
.087	2.21
.199	5.05
.398	10.11
.497	12.62
.676	17.17
.737	18.72
.938	23.83
.991	25.17
1.000	25.40
1.122	28.50
1.125	28.58
1.137	28.88
1.557	39.55
1.875	47.63
1.983	50.37

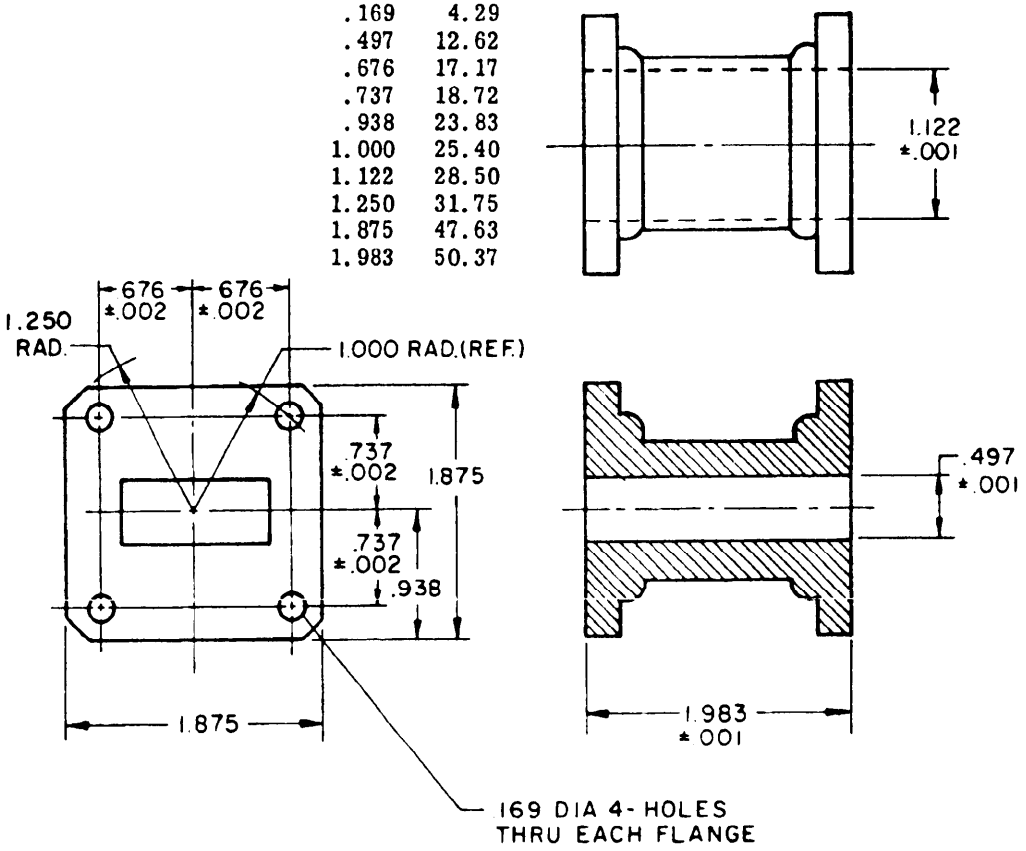


NOTE: Tolerance on all decimal dimensions to be ±.005 unless otherwise specified.

FIGURE F. Differential transmission standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:59Z
Check the source to verify that this is the current version before use.

INCHES	MM
.001	.03
.002	.05
.169	4.29
.497	12.62
.676	17.17
.737	18.72
.938	23.83
1.000	25.40
1.122	28.50
1.250	31.75
1.875	47.63
1.983	50.37



NOTE: Tolerance on all decimal dimensions to be ±.005 unless otherwise specified.

FIGURE G. Precision standard guide.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:59Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding General
U.S.Army Electronics Command
ATTN: AMSEL-RD-TS
Fort Monmouth, New Jersey 07703

FOLD

To detach this form, cut along this line

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:59Z
Check the source to verify that this is the current version before use.

To detach this form, cut along this line

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/1000A ELECTRON TUBE, GAS SWITCHING TYPE 6164

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:59Z
Check the source to verify that this is the current version before use.